

Rajshahi University of Engineering & Technology
Department of Computer Science & Engineering
CSE 3102, Lab No. 10, Lab Exercise

Consider the following instructions. Execute each one and write a report on the answers to the instructions. First complete the following tasks to create the database and the tables.

Database Name: MovieAnalyticsDB

Step 1: Create the following tables with appropriate constraints:

1. Movies

- movie_id (INT, PK, AUTO_INCREMENT, NOT NULL)
- title (VARCHAR(100), NOT NULL)
- release_year (YEAR, NOT NULL)
- genre (VARCHAR(50), NOT NULL)

2. Actors

- actor_id (INT, PK, AUTO_INCREMENT, NOT NULL)
- actor_name (VARCHAR(100), NOT NULL)
- nationality (VARCHAR(50))
- date_of_birth (DATE, NOT NULL)

3. Roles

- role_id (INT, PK, AUTO_INCREMENT, NOT NULL)
- actor_id (INT, FK, NOT NULL)
- movie_id (INT, FK, NOT NULL)
- character_name (VARCHAR(100))
- screen_time_minutes (INT DEFAULT 0)

Step 2: Insert the following sample data:

Movies

title	release_year	genre
----- -----		
Inception	2010	Sci-Fi
The Dark Knight	2008	Action
Interstellar	2014	Sci-Fi
Titanic	1997	Romance
Joker	2019	Drama

Actors

actor_name	nationality	date_of_birth
----- -----		
Leonardo DiCaprio	American	1974-11-11
Christian Bale	British	1974-01-30
Anne Hathaway	American	1982-11-12
Kate Winslet	British	1975-10-05
Joaquin Phoenix	American	1974-10-28

Roles

actor_id	movie_id	character_name	screen_time_minutes
1	1	Dom Cobb	90
1	4	Jack Dawson	80
2	2	Bruce Wayne	110
3	3	Brand	75
4	4	Rose DeWitt Bukater	85
----- ----- -----			

Step 3: Write queries to solve the following:

1. Display all movies.
2. List all actors born after 1980.
3. Find all roles with screen time more than 100 minutes.
4. Count the number of movies in each genre.
5. Show each actor with total screen time.
6. Find actors who acted in more than one movie.
7. Find movies where Leonardo DiCaprio acted.
8. Show average screen time per movie.
9. List top 3 actors with highest total screen time.
10. List genres along with number of unique actors participated.

Database Name : MovieAnalyticsDB

Step 01: CREATE DATABASE MovieAnalyticsDB;

Step 02:

Movie Table:

				movie_id	tilte	release_year	genre
☐	 Edit	 Copy	 Delete	1	Inception	2010	Sci-Fi
☐	 Edit	 Copy	 Delete	2	The Dark Knight	2008	Action
☐	 Edit	 Copy	 Delete	3	Interstellar	2014	Sci-Fi
☐	 Edit	 Copy	 Delete	4	Titanic	1997	Romance
☐	 Edit	 Copy	 Delete	5	Joker	2019	Drama
☐ Check all With selected:  Edit  Copy  Delete  Export							

Actor Table:

←T→				actor_id	actor_name	nationality	date_of_birth
☐  Edit  Copy  Delete	1	Leonardo DiCaprio	American	1974-11-11			
☐  Edit  Copy  Delete	2	Christian Bale	British	1974-01-30			
☐  Edit  Copy  Delete	3	Anne Hathaway	American	1982-11-12			
☐  Edit  Copy  Delete	4	Kate Winslet	British	1975-10-05			
☐  Edit  Copy  Delete	5	Joaquin Phoenix	American	1974-10-28			
↑ ☐ Check all With selected:  Edit  Copy  Delete							

Roles Table:

			role_id	actor_id	movie_id	character_name	screen_time_minutes
☐	Edit	Copy	Delete	1	1	1 Dom Cobb	90
☐	Edit	Copy	Delete	2	1	4 Jack Dawson	80
☐	Edit	Copy	Delete	3	2	2 Bruce Wayne	110
☐	Edit	Copy	Delete	4	3	3 Brand	75
☐	Edit	Copy	Delete	5	4	4 Rose DeWitt Bukater	85
☐	Edit	Copy	Delete	6	5	5 Arthur Fleck	115

☐ Check all With selected: Edit Copy Delete Export

Step 03:

1. Display All movies

SELECT * FROM movies

				movie_id	title	release_year	genre
☐	Edit	Copy	Delete	1	Inception	2010	Sci-Fi
☐	Edit	Copy	Delete	2	The Dark Knight	2008	Action
☐	Edit	Copy	Delete	3	Interstellar	2014	Sci-Fi
☐	Edit	Copy	Delete	4	Titanic	1997	Romance
☐	Edit	Copy	Delete	5	Joker	2019	Drama

☐ Check all With selected: Edit Copy Delete Export

2. List all actors born after 1980

SELECT * FROM `actors`
WHERE date_of_birth > '1980-12-31';

				actor_id	actor_name	nationality	date_of_birth
☐	Edit	Copy	Delete	3	Anne Hathaway	American	1982-11-12

☐ Check all With selected: Edit Copy Delete Export

3. Find all roles with screen time more than 100 minutes.

SELECT * FROM `roles`
WHERE screen_time_minutes >= '100';

				role_id	actor_id	movie_id	character_name	screen_time_minutes
☐	Edit	Copy	Delete	3	2	2	Bruce Wayne	110
☐	Edit	Copy	Delete	6	5	5	Arthur Fleck	115

4. Count the number of movies in each genre.

```
SELECT genre, COUNT(*) as movie_count
FROM `movies`
GROUP BY genre;
```

genre	movie_count
Action	1
Drama	1
Romance	1
Sci-Fi	2

5. Show each actor with total screen time.

```
SELECT A.actor_name, SUM(R.screen_time_minutes) AS total_screen_time
FROM Actors A
JOIN Roles R
ON A.actor_id = R.actor_id
GROUP BY A.actor_id, A.actor_name;
```

actor_name	total_screen_time
Leonardo DiCaprio	170
Christian Bale	110
Anne Hathaway	75
Kate Winslet	85
Joaquin Phoenix	115

6. Find actors who acted in more than one movie.

```
SELECT A.actor_name, COUNT(DISTINCT R.movie_id) AS movie_count
FROM Actors A
JOIN Roles R
ON A.actor_id = R.actor_id
GROUP BY A.actor_id, A.actor_name
HAVING movie_count > 1;
```

actor_name	movie_count
Leonardo DiCaprio	2

7. Find movies where Leonardo DiCaprio acted.

```
SELECT M.title
FROM Movies M
```

```

JOIN Roles R ON M.movie_id = R.movie_id
JOIN Actors A ON R.actor_id = A.actor_id
WHERE A.actor_name = 'Leonardo DiCaprio';

```

title

Inception

Titanic

8. Show average screen time per movie
- ```

SELECT M.title, AVG(R.screen_time_minutes) AS avg_screen_time
FROM Movies M
JOIN Roles R ON M.movie_id = R.movie_id
GROUP BY M.movie_id, M.title;

```

| title           | avg_screen_time |
|-----------------|-----------------|
| Inception       | 90.0000         |
| The Dark Knight | 110.0000        |
| Interstellar    | 75.0000         |
| Titanic         | 82.5000         |
| Joker           | 115.0000        |

9. List top 3 actors with highest total screen time.

```

SELECT A.actor_name, SUM(screen_time_minutes) as total_screen_time
FROM actors A
JOIN roles R
ON A.actor_id = R.actor_id
GROUP BY A.actor_id, A.actor_name
ORDER by total_screen_time DESC
LIMIT 3;

```

| actor_name        | total_screen_time | ▼ 1 |
|-------------------|-------------------|-----|
| Leonardo DiCaprio | 170               |     |
| Joaquin Phoenix   | 115               |     |
| Christian Bale    | 110               |     |

10. List genres along with number of unique actors participated

```

SELECT M.genre, COUNT(DISTINCT R.actor_id) AS unique_actor_count
FROM Movies M
JOIN Roles R ON M.movie_id = R.movie_id
GROUP BY M.genre;

```

| genre   | unique_actor_count |
|---------|--------------------|
| Action  | 1                  |
| Drama   | 1                  |
| Romance | 2                  |
| Sci-Fi  | 2                  |